

Generalization of the Limit

Let us define a generalized form of the limit as:

$$\lim_{n \rightarrow \infty} n^a \left((n^b + c)^{1/k} - n^{b/k} \right)$$

where $a, b, c, k \in \mathbb{R}$, with $k > 0$ and $b > 0$.

To find a non-zero, finite limit, the powers of n must precisely balance out. We can prove this using the generalized binomial theorem (or Taylor expansion) for large n .

Rigorous Derivation

1. Factor out the dominant term

First, we factor out n^b from inside the k -th root:

$$(n^b + c)^{1/k} = \left(n^b \left(1 + \frac{c}{n^b} \right) \right)^{1/k} = n^{b/k} \left(1 + \frac{c}{n^b} \right)^{1/k}$$

2. Apply the Binomial Expansion

Because $n \rightarrow \infty$ and $b > 0$, the term $\frac{c}{n^b} \rightarrow 0$. We can use the binomial expansion $(1 + x)^p = 1 + px + \mathcal{O}(x^2)$ for small x :

$$\left(1 + \frac{c}{n^b} \right)^{1/k} = 1 + \frac{c}{kn^b} + \mathcal{O}\left(\frac{1}{n^{2b}}\right)$$

Substituting this back into our factored expression:

$$\begin{aligned} (n^b + c)^{1/k} &= n^{b/k} \left(1 + \frac{c}{kn^b} + \mathcal{O}\left(\frac{1}{n^{2b}}\right) \right) \\ &= n^{b/k} + \frac{c}{k} n^{b/k-b} + \mathcal{O}\left(n^{b/k-2b}\right) \end{aligned}$$

3. Substitute back into the Limit

Now, we place this expansion back into the original generalized limit:

$$\lim_{n \rightarrow \infty} n^a \left(\left[n^{b/k} + \frac{c}{k} n^{b/k-b} + \mathcal{O}\left(n^{b/k-2b}\right) \right] - n^{b/k} \right)$$

The $n^{b/k}$ terms cancel out, leaving:

$$\lim_{n \rightarrow \infty} n^a \left(\frac{c}{k} n^{b/k-b} + \mathcal{O}\left(n^{b/k-2b}\right) \right) = \lim_{n \rightarrow \infty} \left(\frac{c}{k} n^{a+b/k-b} + \mathcal{O}\left(n^{a+b/k-2b}\right) \right)$$

4. Establish Convergence Conditions

For this limit to converge to a non-zero, finite constant, the leading exponent of n must be equal to 0:

$$a + \frac{b}{k} - b = 0 \implies a = b \left(1 - \frac{1}{k} \right)$$

If this condition holds, the limit evaluates directly to the leading coefficient:

$$L = \frac{c}{k}$$

Application to the Original Problem

For the specific limit:

$$\lim_{n \rightarrow \infty} n^{3/2} \left(\sqrt{n^3 + 1} - n\sqrt{n} \right)$$

We can map this to our generalized variables:

- $a = 3/2$
- $b = 3$
- $c = 1$
- $k = 2$ (since it is a square root)

Let us check if the convergence condition $a = b(1 - 1/k)$ is met:

$$\frac{3}{2} = 3 \left(1 - \frac{1}{2} \right) \implies \frac{3}{2} = \frac{3}{2}$$

Because the condition holds perfectly, we can apply the generalized result $L = \frac{c}{k}$:

$$L = \frac{1}{2}$$